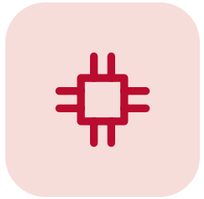
Meet AI

What AI actually is, the tools you can use, and how to get them

Graham Pellegrini

Lecture 1 · Wednesday 28 October 2026

Who am I?



Computer Engineering

B.Sc. (Hons), University of Malta

→



Master of AI

University of Malta, full-time research

→



AI researcher

Research Support Officer, Project M.AI.ESTRO

→



PhD

next step

AI helped build these slides. **I made every decision.**

How this course works



Bring a laptop
every week



eduroam Wi-Fi
setup guide



Attendance
the only assessment



Wednesdays
12:00 to 14:00, break at 13:00



On the VLE
slides and notes after class



Ask anything
there is no fixed syllabus

Eight Wednesdays

28 Oct	**Meet AI**	what it is, the tools, access
04 Nov	**Tokens and better results**	how to ask well
11 Nov	**Research and documents**	search, summarise, check
18 Nov	**Writing and communication**	emails, reports, messages
25 Nov	**Planning and your own assistant**	plans, minutes, Gems
02 Dec*	**Presentations and visuals**	slides, notes, images
09 Dec*	**Tables and data**	spreadsheets, no coding
16 Dec*	**Your AI workflow**	a process for your field

* December dates to be confirmed

At the end of this course, you shall be able to

1. Identify common types of AI tools and select an appropriate tool for a given task
2. Give clear instructions by defining the purpose, context, audience, constraints and required format
3. Use AI to support research, document work, professional communication, planning, presentations and simple data interpretation
4. Compare, refine and combine AI-generated outputs rather than accepting the first response
5. Create reusable prompts and practical workflows relevant to your intended profession
6. Recognise when an output requires verification, expert advice or a human decision

Understanding of the class

A short survey, about 5 minutes. No wrong answers.

Scan and fill in the survey

QR code to the Google Form

[add QR code and short link]

- About 5 minutes
- Anonymous
- Some questions are just for fun

What AI actually is

Let's get a common understanding. No maths, no coding.

Common myths

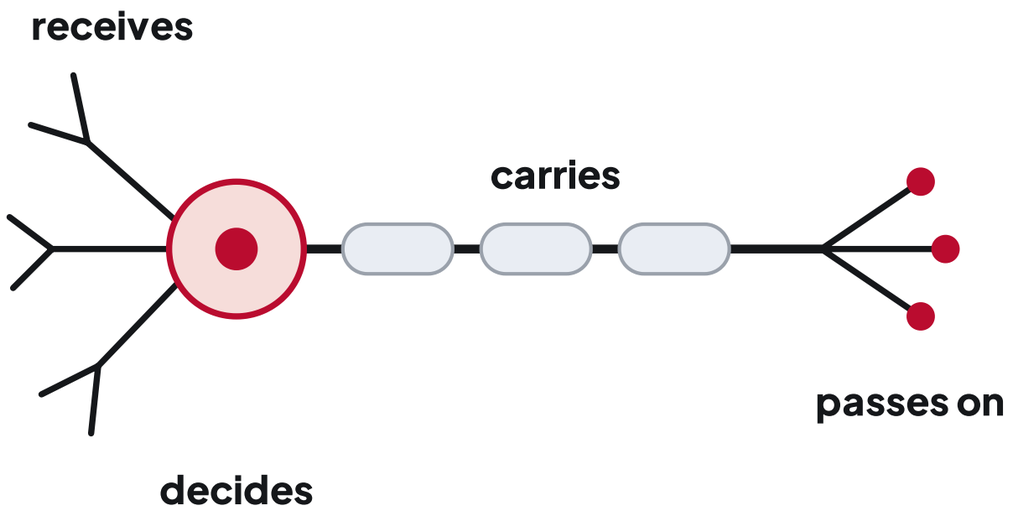
MYTH	REALITY
~~It knows everything~~	→ It learned patterns from text
~~It looks the answer up~~	→ It writes from what it learned
~~It thinks like a person~~	→ Inspired by the brain, but it predicts likely words

More common myths

MYTH		REALITY
~~It is always right~~	→	It sounds confident even when it is wrong
~~It is always up to date~~	→	It knows what it was trained on, unless it searches
~~It is neutral~~	→	It reflects the text it learned from, biases included

AI chatbots learn patterns from huge amounts of text, then predict what comes next.

Start with your brain



86 billion

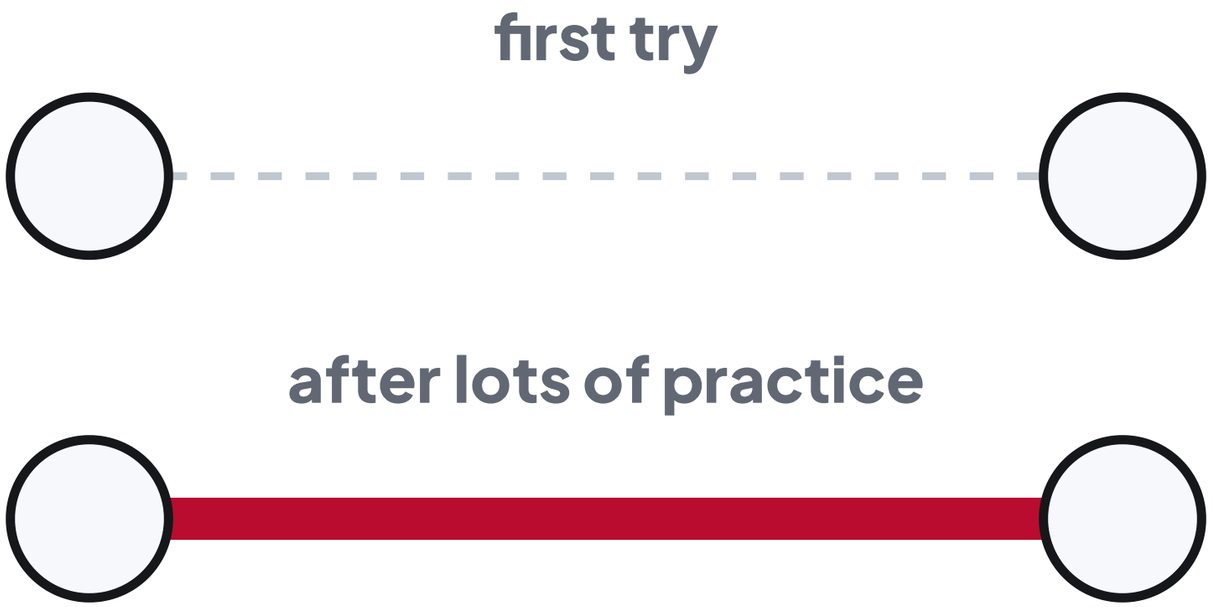
neurons in your brain

thousands

of connections each

Azevedo et al. (2009), Journal of Comparative Neurology

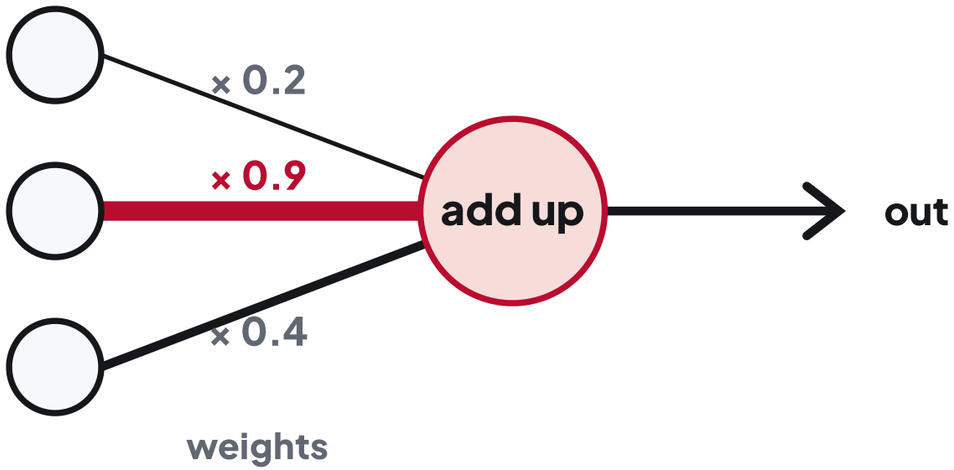
Learning makes connections stronger



Neurons that fire together, wire together

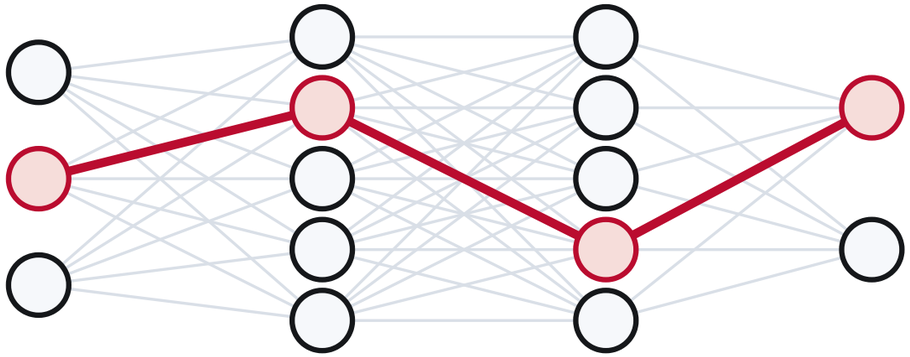
After Donald Hebb (1949)

An artificial neuron is a simple maths copy



- Numbers come in
- Each connection has a strength (a “weight”)
- Add up and pass on

Connect lots of them: a neural network



Billions

of connections inside a chatbot

Trillions

of connections in your brain

Brain figure is a common estimate

How it learns: guess, check, adjust



Hide the next word

"The cat sat on the …"

→



Guess

"floor"

→



Compare

the real word was "mat"

→



Adjust the connections

"mat" a little more likely

Repeat trillions of times. Nobody writes the rules.

A bit like a child learning a language

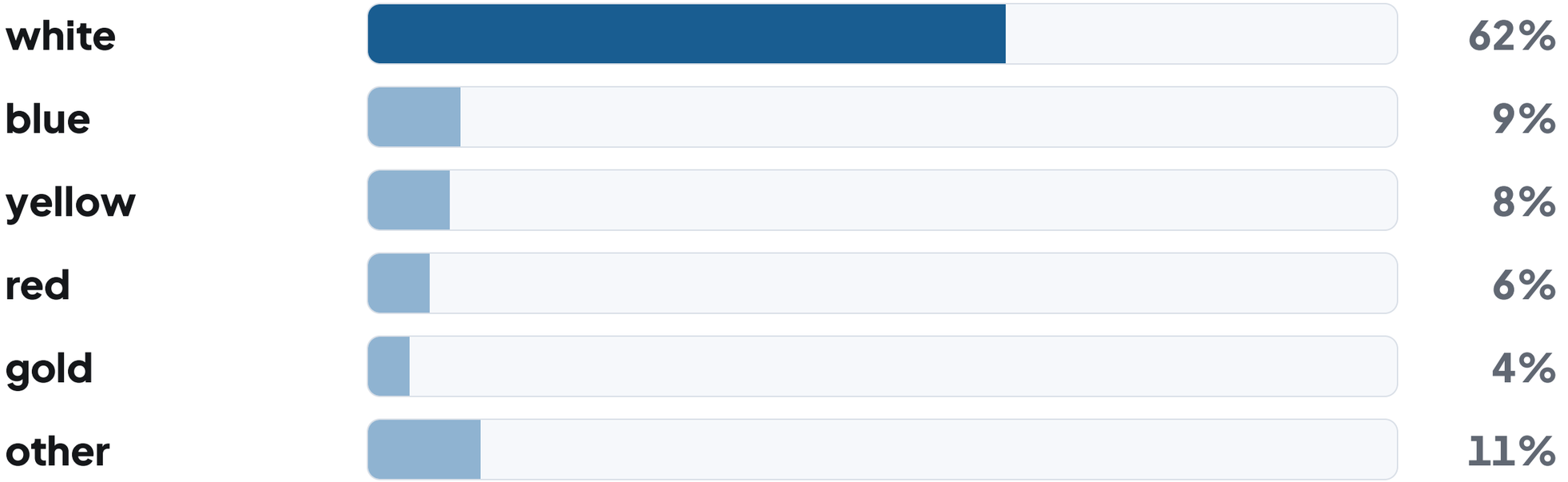
	A CHILD	AN AI MODEL
Learns from	Thousands of sentences	Trillions of words
Grammar rules	Not taught first	Never programmed
Ends up able to	Communicate with people	Predict the next word
Experiences the world	**Yes**	**No, only text**

Finish the phrase

Shout out the first word that comes to mind

Black and _____

Black and …

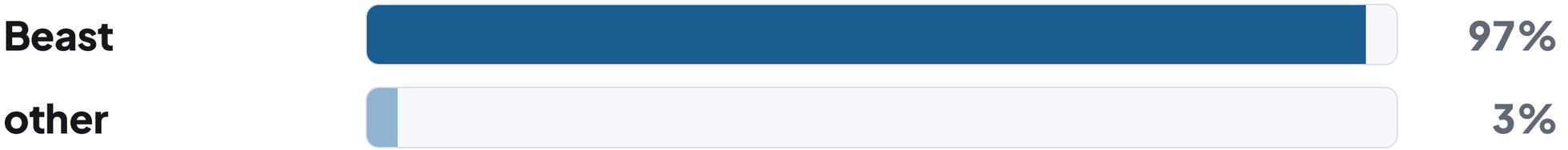


Estimated by an AI model (Claude Sonnet 5.5), typical values

Your brain picked the most familiar ending. That is a prediction.

Beauty and the _____

Beauty and the …

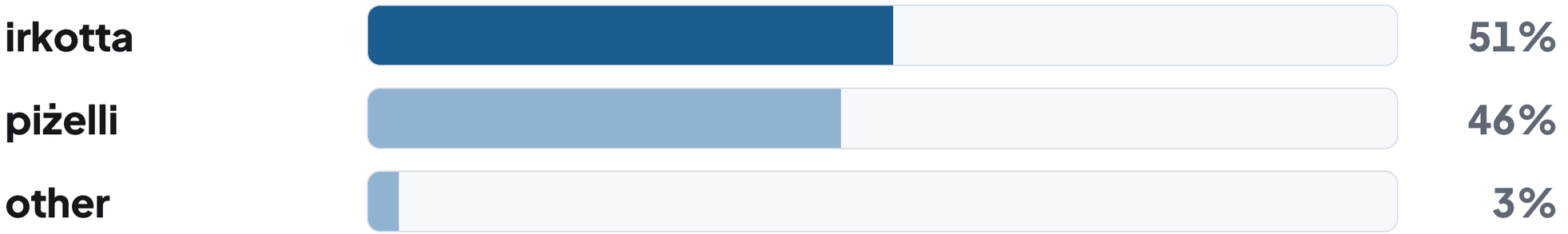


Estimated by an AI model (Claude Sonnet 5.5), typical values

One ending dominates, so the prediction is almost certain.

Pastizzi tal-

Pastizzi tal- …



Illustrative numbers

Almost a tie, so ask twice and you may get a different answer.

The capital of Malta is

The capital of Malta is …

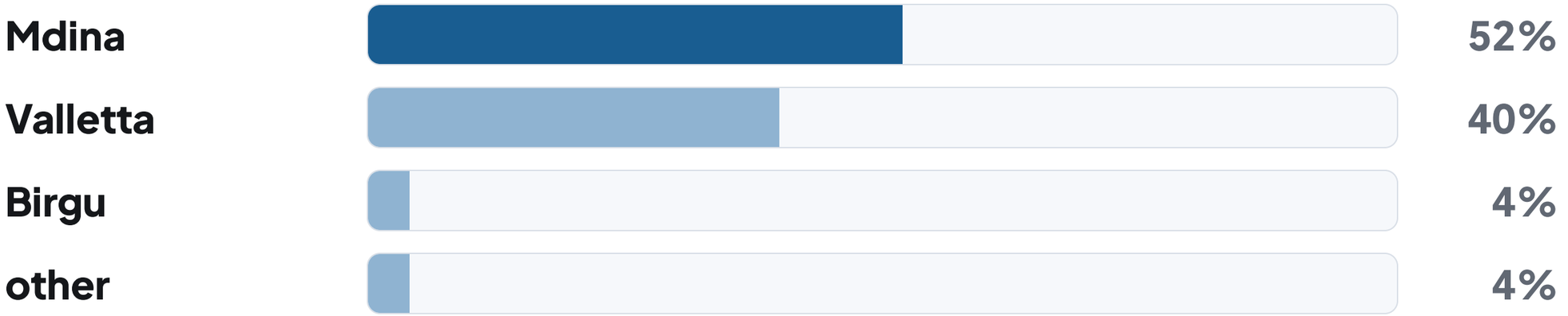


Estimated by an AI model (Claude Sonnet 5.5), typical values

Common facts come out right, but they are still predictions.

The capital of Malta was

The capital of Malta was …

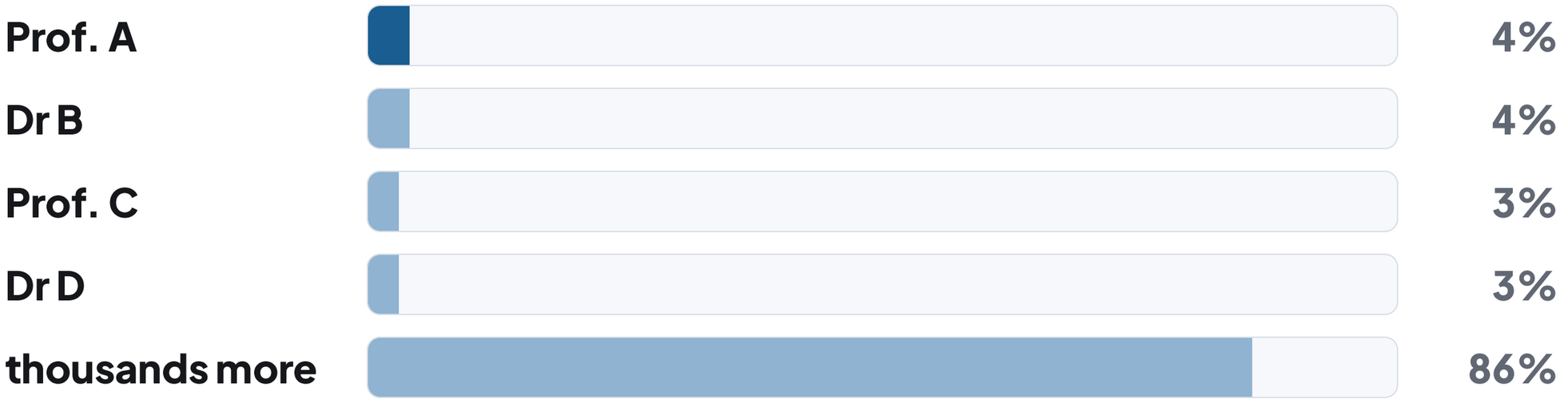


Estimated by an AI model (Claude Sonnet 5.5), typical values

One word changed, and the prediction moved to the past.

The best lecturer at UM is

The best lecturer at UM is …



Illustrative numbers

No real answer, but it will still sound confident. This is where made-up answers come from.

What the game shows



Your brain predicts

every phrase, without thinking

→



AI predicts at scale

one piece at a time, again and again

→



Likely is not true

right for common facts, wrong for rare ones

How much did it read?

15 trillion

pieces of text in Meta's Llama 3

85,000 years

for a person reading non-stop

Meta AI, Llama 3 announcement (April 2024). Reading time: 250 words a minute, 24 hours a day, about 0.75 words per piece.

Why it sounds so sure of itself

✓ **Trained to write text that sounds right**

✗ **Not built to know whether it is true**

Great at drafting and explaining. **You check the facts.**

It does not read words, it reads tokens

English: 9 tokens

`Un` `bel` `ievable` ` results` ` at` ` the` ` University` ` of` ` Malta`

Maltese: 2 words, 6 tokens

`Past` `izzi` ` tal` `-` `irk` `otta`

Split with OpenAI's o200k tokenizer, used by recent ChatGPT models

Part A in three sentences

1. AI predicts what comes next
2. What it learned is stored in connection strengths, not facts
3. Likely is not true: you check and decide

5 min

Back at 13:05 for the tools, and how to get them

The tools, and how to get them

The big three, staying safe, and free access

The big three



ChatGPT

by OpenAI



Gemini

by Google



Claude

by Anthropic

Plenty of other names, often the same engines

Microsoft Copilot · runs on OpenAI

Meta AI · in WhatsApp

Perplexity · AI search

Mistral · European

DeepSeek · Chinese

Thousands of apps · built on top

You do not need to pay for anything

€0

free plans: enough for this course

€5 to €25

a month for paid plans

€0

paid features through UM, AI for All, Revolut

Official pricing pages, October 2026. Prices change often.

Same question, three assistants

```
I am a first-year student at the University of Malta.
Give me 5 short tips on how I could use AI to prepare for an exam,
and 1 thing I should be careful about.
```

Which answer did you prefer?

ChatGPT	0
Gemini	0
Claude	0
No real difference	0

There is no single "AI answer". Compare, combine, and then you decide.

Two kinds of AI tool

Chat assistant · ChatGPT, Gemini, Claude



You ask → It answers → You decide what to do

Agent · Claude Code, OpenAI Codex, ChatGPT agent



You set a goal → It opens files and apps → It works step by step → It reports back

What they look like

Chat assistant

Can you make my email to my lecturer sound more polite?

Of course. Here is a more polite version: “Dear Dr Borg, I hope you are well…”

Shorter please.

Agent on your computer

```
> Tidy my thesis folder and fix the references
Reading 14 files…
✓ Renamed 6 chapter files
✓ Fixed 23 references
? Delete old_draft.docx? Allow / Deny
```

Illustrations, not real screenshots

In this course we use chat assistants

- ✓ Free and safe to learn with
- ✓ Everything you learn carries over to agents
- ✓ Agents come back in Lecture 8

What can AI tools do for you?

Chat and writing
explain, draft, rewrite

Search and research
sources with links

Your documents
ask about your own PDFs

Images and slides
visual ideas, outlines

Meetings and audio
transcripts, minutes

Tables and data
questions, patterns, charts

Which type of tool would you use?

1. Summarise a 40-page reading
2. Find recent research on nurse burnout
3. Turn notes into a polite email

Which type of tool would you use?

Summarise a 40-page reading	→ Your documents
Find recent research on nurse burnout	→ Search and research
Turn notes into a polite email	→ Chat and writing

Stay in charge

When to check, what not to share, and the settings that protect you

Check it, ask an expert, or decide yourself

Facts, numbers, dates, references	→ Check it
Health, legal, money, safety	→ Ask an expert
Judgement calls, decisions about people	→ Decide yourself

What not to paste into an AI tool

- ✕ Other people's personal details
- ✕ Passwords, ID numbers, bank details
- ✕ Confidential work or unpublished material

Would you pin it on a public noticeboard?

AI in assessed work

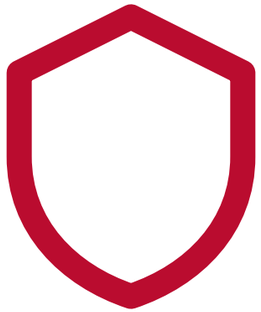
- Check each assignment's rules
- Be open about how you used AI
- You own every mistake you submit

UM guidance "Using Generative AI in student work", on the VLE

Your data, your settings

TOOL	STOP TRAINING ON YOUR CHATS	PRIVATE CHAT
ChatGPT	Settings → **Data controls** → **Improve the model** off	**Temporary**
Gemini (personal)	Gemini Apps Activity → **Keep Activity** off	**Temporary**
Gemini (UM)	Nothing to do: never used for training	
Claude	Settings → **Privacy** → model improvement off	**Incognito**

OpenAI Data Controls FAQ; Gemini Apps Privacy Hub; Anthropic Privacy Center; UM IT Services (October 2026)



With your UM account, Gemini does not train on your chats and no human reviews them

UM IT Services, um.edu.mt/itservices/google

Your setup for this week

Two essentials, then two optional extras

Essential 1: Gemini with your UM account

1 → **2** → **3**

gemini.google.com

Sign in with UM

not personal Gmail

Check the corner

shows your UM account

Deep Research | Gems | Canvas | Audio Overviews | Gemini Live | Gemini Notebook

Essential 2: a free ChatGPT account

1 → 2 → 3

chatgpt.com

Sign up

Email or Google

Microsoft or Apple also work

Turn off training

Settings → Data controls

The free plan is enough for this course

Optional: AI for All (AI Għal Kulħadd)

Free

national AI course by UM and MDIA

3 modules

online, at your own pace

1 exam

then your reward

ai4all.gov.mt

Who can join, and what you get

Who

- ✓ Maltese citizens and residents
- ✓ Aged 14 or over
- ✓ With an active e-ID

ai4all.gov.mt/faqs. Conditions apply.

What you get

- ✓ **1 year free ChatGPT Plus**
  or Microsoft 365 with Copilot
- ✓ **A verifiable certificate**

How to sign up

1 → **2** → **3**

Register with e-ID
ai4all.gov.mt

Onboarding form
accept the terms

Activate the bundle
My Catalogues

4 → **5** → **6**

Do 3 modules
on Coursera

Final exam
on the platform

Code by email
and in My Profile

Tips for AI for All

- Check your e-ID email works: the code goes there
- Pick ChatGPT Plus as your reward
- Not eligible? Free plans are enough

Help: Help & Support on the platform dashboard, or servizz@gov.mt

Already with Revolut?

PLAN	INCLUDES
Revolut Premium	ChatGPT Go for **6 months**
Revolut Metal	ChatGPT Go for **12 months**

Not a sponsor. Do not sign up just for this, but if you already pay, claim it.

revolut.com/en-MT plan pages, October 2026. Replaced Perplexity Pro in September 2026.



Optional: try Claude

Free at claude.ai. Good for comparing answers.

Before next Wednesday

- [ ] Open Gemini with your UM account
- [ ] Create a free ChatGPT account and turn off training
- [ ] Send the next prompt to both, save the answers
- [ ] Optional: start AI for All, claim Revolut

Your first real prompt

```
I am studying [your course] at the University of Malta.
Suggest three ways AI could help me with my studies this semester,
and one way it could cause me problems.
```

Which answer was more useful, and why? Bring one observation.

Today in six points

1. AI predicts what comes next
2. Likely is not true: check facts
3. Compare ChatGPT, Gemini, Claude
4. Keep personal data out; use UM for uni work
5. Set up Gemini and ChatGPT this week
6. AI drafts, you decide

Questions?

Next week, 4 November: **tokens and better results**